MA1522 Linear Algebra for Computing

Lesson 14: Orthogonal Projection and the Gram-Schmidt Process

Announcement: Online Quiz 3

- **Quiz 3 covers Chapter 3**, Sections 3.1–3.8.

- It opens tomorrow, **Saturday 3 October, 0:00**, and is due **Sunday 4 October, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson14` (e.g. `A1234567Z_lesson14.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 14**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Example 1

True or false? Let V be a subspace of $\mathbb{R}^n$ and $\mathbf{w} \in \mathbb{R}^n$. If

$$\mathbf{w} = \mathbf{v}_p + \mathbf{v}_n = \mathbf{u}_p + \mathbf{u}_n, \qquad \text{where } \mathbf{v}_p, \mathbf{u}_p \in V \text{ and } \mathbf{v}_n, \mathbf{u}_n \in V^\perp,$$

then

1. $\mathbf{v}_p = \mathbf{u}_p$;
2. $\mathbf{v}_n = \mathbf{u}_n$.

1 and 2 are both true. Let $\mathbf{d} = \mathbf{v}_p - \mathbf{u}_p = \mathbf{u}_n - \mathbf{v}_n$. Then $\mathbf{d} \in V$ and $\mathbf{d} \in V^\perp$, since both are subspaces. So $\mathbf{d}$ is orthogonal to itself: $\mathbf{d} \cdot \mathbf{d} = 0$, hence $\mathbf{d} = \mathbf{0}$, that is, $\mathbf{v}_p = \mathbf{u}_p$ and $\mathbf{v}_n = \mathbf{u}_n$.

Section 5.3 Orthogonal Projection

Let V be a subspace of $\mathbb{R}^n$.

Theorem (Orthogonal projection theorem)

Every vector $\mathbf{w}$ in $\mathbb{R}^n$ can be decomposed uniquely as a sum $\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$, where $\mathbf{w}_p \in V$ and $\mathbf{w}_n \perp V$. Moreover, if $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is an orthogonal basis for V, then

$$\mathbf{w}_p = \frac{\mathbf{w} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{w} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2 + \cdots + \frac{\mathbf{w} \cdot \mathbf{u}_k}{\mathbf{u}_k \cdot \mathbf{u}_k}\mathbf{u}_k.$$

Definition

$\mathbf{w}_p$ is the **orthogonal projection** (or just **projection**) of $\mathbf{w}$ onto V.

Theorem (Best approximation theorem)

$\mathbf{w}_p$ is the vector in V closest to $\mathbf{w}$: $\|\mathbf{w} - \mathbf{w}_p\| \leq \|\mathbf{w} - \mathbf{v}\|$ for all $\mathbf{v} \in V$.

Question 1

Let $V = \text{span}\{\mathbf{a}_1, \mathbf{a}_2\} \subseteq \mathbb{R}^3$ and $\mathbf{w} \in \mathbb{R}^3$, where

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 5 \\ -3 \\ 7 \end{pmatrix}.$$

1. Write $\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$, where $\mathbf{w}_p \in V$ and $\mathbf{w}_n \perp V$.

2. Find the smallest possible value of $\|\mathbf{w} - \mathbf{v}\|$ for $\mathbf{v} \in V$, and the $\mathbf{v}$ that attains it.

Question 1 — solution

1. First, $\mathbf{a}_1 \cdot \mathbf{a}_2 = 1 + 3 - 4 = 0$, so $\{\mathbf{a}_1, \mathbf{a}_2\}$ is an orthogonal basis for V and the projection formula applies:

$$\mathbf{w}_p = \frac{\mathbf{w} \cdot \mathbf{a}_1}{\mathbf{a}_1 \cdot \mathbf{a}_1}\mathbf{a}_1 + \frac{\mathbf{w} \cdot \mathbf{a}_2}{\mathbf{a}_2 \cdot \mathbf{a}_2}\mathbf{a}_2 = \frac{28}{14}\mathbf{a}_1 + \frac{-6}{6}\mathbf{a}_2 = \begin{pmatrix} 1 \\ -5 \\ 6 \end{pmatrix}, \qquad \mathbf{w}_n = \mathbf{w} - \mathbf{w}_p = \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}.$$

$\mathbf{w}_n \cdot \mathbf{a}_1 = 4 - 6 + 2 = 0$ and $\mathbf{w}_n \cdot \mathbf{a}_2 = 4 - 2 - 2 = 0$. ✓

2. By the best approximation theorem, the closest vector in V is $\mathbf{v} = \mathbf{w}_p = (1, -5, 6)^T$, and the smallest value is

$$\|\mathbf{w} - \mathbf{w}_p\| = \|\mathbf{w}_n\| = \sqrt{16 + 4 + 1} = \sqrt{21}. \qquad \|\mathbf{w}\|^2 = 83 = 62 + 21 = \|\mathbf{w}_p\|^2 + \|\mathbf{w}_n\|^2. \checkmark$$

This is the distance from $\mathbf{w}$ to the plane V. □

Section 5.3 Gram-Schmidt Process

Gram-Schmidt process

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a linearly independent set. Let

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{u}_1, \\
\mathbf{v}_2 &= \mathbf{u}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{u}_2}{\|\mathbf{v}_1\|^2}\mathbf{v}_1, \\
\mathbf{v}_3 &= \mathbf{u}_3 - \frac{\mathbf{v}_1 \cdot \mathbf{u}_3}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_3}{\|\mathbf{v}_2\|^2}\mathbf{v}_2, \\
&\vdots \\
\mathbf{v}_k &= \mathbf{u}_k - \frac{\mathbf{v}_1 \cdot \mathbf{u}_k}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \cdots - \frac{\mathbf{v}_{k-1} \cdot \mathbf{u}_k}{\|\mathbf{v}_{k-1}\|^2}\mathbf{v}_{k-1}.
\end{aligned}$$

Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is an orthogonal set of nonzero vectors with $\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$, and $\left\{\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \ldots, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}\right\}$ is an orthonormal basis for it.

If S is not linearly independent, omit every $\mathbf{v}_i$ that is $\mathbf{0}$ (and leave it out of the later steps).

Question 2

Let $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\} \subseteq \mathbb{R}^4$, where

$$\mathbf{u}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix}, \qquad \mathbf{u}_2 = \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix}, \qquad \mathbf{u}_3 = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix}, \qquad \mathbf{u}_4 = \begin{pmatrix} 1 \\ 1 \\ -4 \\ 3 \end{pmatrix}.$$

Apply the Gram-Schmidt process to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ to obtain an orthogonal basis for V.

Question 2 — solution: $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$

Apply the process, one vector at a time. First, $\mathbf{v}_1 = \mathbf{u}_1 = (2, 1, 0, 2)^T$ and $\|\mathbf{v}_1\|^2 = 4 + 1 + 0 + 4 = 9$.

$\mathbf{v}_2$: $\mathbf{v}_1 \cdot \mathbf{u}_2 = 6 - 1 + 0 + 4 = 9$, so

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{u}_2}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 = \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix} - \frac{9}{9}\begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix}, \qquad \|\mathbf{v}_2\|^2 = 1 + 4 + 4 + 0 = 9.$$

$\mathbf{v}_3$: $\mathbf{v}_1 \cdot \mathbf{u}_3 = 2 + 3 + 0 + 4 = 9$ and $\mathbf{v}_2 \cdot \mathbf{u}_3 = 1 - 6 - 4 + 0 = -9$, so

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{v}_1 \cdot \mathbf{u}_3}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_3}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix} - \frac{9}{9}\begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} - \frac{-9}{9}\begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

So $\mathbf{u}_3 \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$: discard $\mathbf{v}_3$ and continue with $\mathbf{v}_1, \mathbf{v}_2$ only.

Question 2 — solution: $\mathbf{v}_4$

$\mathbf{v}_4$: $\mathbf{v}_1 \cdot \mathbf{u}_4 = 2 + 1 + 0 + 6 = 9$ and $\mathbf{v}_2 \cdot \mathbf{u}_4 = 1 - 2 - 8 + 0 = -9$, so

$$\mathbf{v}_4 = \mathbf{u}_4 - \frac{\mathbf{v}_1 \cdot \mathbf{u}_4}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_4}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -4 \\ 3 \end{pmatrix} - \frac{9}{9}\begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} - \frac{-9}{9}\begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix}.$$

$\mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_1 \cdot \mathbf{v}_4 = \mathbf{v}_2 \cdot \mathbf{v}_4 = 0$. ✓

An orthogonal basis for V is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\} = \{(2, 1, 0, 2)^T, (1, -2, 2, 0)^T, (0, -2, -2, 1)^T\}$, so $\dim(V) = 3$. □

Question 3

Let V and $\mathbf{u}_1, \ldots, \mathbf{u}_4$ be as in Question 2, and let $\mathbf{w} = \begin{pmatrix} 3 \\ -1 \\ -7 \\ 2 \end{pmatrix}$.

1. Using the orthogonal basis for V you found in Question 2, find the projection $\mathbf{w}_p$ of $\mathbf{w}$ onto V.

2. The set $T = \left\{ \mathbf{u}_2, \ \mathbf{u}_3, \ \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \right\}$ is another orthogonal basis for V. Use T to compute the projection of $\mathbf{w}$ onto V again, and compare with part 1.

Question 3 — solution: part 1

1. By Question 2, $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is an orthogonal basis for V, with $\|\mathbf{v}_1\|^2 = \|\mathbf{v}_2\|^2 = \|\mathbf{v}_4\|^2 = 9$. Compute the inner products with $\mathbf{w}$:

$$\mathbf{w} \cdot \mathbf{v}_1 = 6 - 1 + 0 + 4 = 9, \quad \mathbf{w} \cdot \mathbf{v}_2 = 3 + 2 - 14 + 0 = -9, \quad \mathbf{w} \cdot \mathbf{v}_4 = 0 + 2 + 14 + 2 = 18.$$

By the orthogonal projection theorem,

$$\mathbf{w}_p = \frac{\mathbf{w} \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \frac{\mathbf{w} \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 + \frac{\mathbf{w} \cdot \mathbf{v}_4}{\|\mathbf{v}_4\|^2}\mathbf{v}_4 = \frac{9}{9}\begin{pmatrix}2\\1\\0\\2\end{pmatrix} + \frac{-9}{9}\begin{pmatrix}1\\-2\\2\\0\end{pmatrix} + \frac{18}{9}\begin{pmatrix}0\\-2\\-2\\1\end{pmatrix} = \begin{pmatrix}1\\-1\\-6\\4\end{pmatrix}.$$

$\mathbf{w}_n = \mathbf{w} - \mathbf{w}_p = (2, 0, -1, -2)^T$, and $\mathbf{w}_n \cdot \mathbf{v}_1 = \mathbf{w}_n \cdot \mathbf{v}_2 = \mathbf{w}_n \cdot \mathbf{v}_4 = 0$. ✓

Question 3 — solution: part 2

2. Write $\mathbf{t} = (0, -2, -2, 1)^T$, so $T = \{\mathbf{u}_2, \mathbf{u}_3, \mathbf{t}\}$. The squared norms and the inner products with $\mathbf{w}$ are

$$\|\mathbf{u}_2\|^2 = 9 + 1 + 4 + 4 = 18, \qquad \mathbf{w} \cdot \mathbf{u}_2 = 9 + 1 - 14 + 4 = 0,$$
$$\|\mathbf{u}_3\|^2 = 1 + 9 + 4 + 4 = 18, \qquad \mathbf{w} \cdot \mathbf{u}_3 = 3 - 3 + 14 + 4 = 18,$$
$$\|\mathbf{t}\|^2 = 0 + 4 + 4 + 1 = 9, \qquad \mathbf{w} \cdot \mathbf{t} = 0 + 2 + 14 + 2 = 18.$$

By the orthogonal projection theorem, now with the basis T,

$$\mathbf{w}_p = \frac{0}{18}\begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix} + \frac{18}{18}\begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix} + \frac{18}{9}\begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -6 \\ 4 \end{pmatrix}.$$

The coefficients differ ($1, -1, 2$ in part 1; $0, 1, 2$ here), but the projection is the same vector: $\mathbf{w}_p$ depends only on V, not on the orthogonal basis. □

Next lesson

Next lesson (**Tue 6 Oct**, Sections 5.4 & 5.5), I will focus on

1. the QR factorization of a matrix with linearly independent columns, and how to compute it by the Gram-Schmidt process;

2. the definition of a least squares solution of a linear system $A\mathbf{x} = \mathbf{b}$, and how to compute it.

Please watch the video clips beforehand.